

DATA SCIENCE & ANALYTICS INTERNSHIP

TASK 1 : BUSINESS SALES PERFORMANCE ANALYSIS

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Internship : Future Interns

Domain : Data Science & Analytics

1.Introduction

The objective of this project is to analyze business sales data and identify important patterns in sales performance. The analysis focuses on revenue trends, top-selling products, high-value categories, and regional performance. Python and Jupyter Notebook were used for data preprocessing, analysis, and visualization. The analysis was performed to generate meaningful business insights that can support better sales planning and decision-making.

Dataset Description

Dataset name: Superstore Dataset

Source: Kaggle

Domain: Retail / Sales Analytics

Number of records: 9,994 rows and 21 columns

Purpose: To analyze sales, profit, customer segments, product categories, and regional performance.

Problem Statement:

Retail businesses generate large amounts of sales data from different customers, products, regions, and time periods. Analyzing this data manually can make it difficult to identify important sales patterns and business performance. Therefore, there is a need for an effective data analytics approach to examine the sales data, identify trends, understand profitable products and categories, and compare regional performance. This project addresses this problem by analyzing the Superstore dataset and presenting the findings through meaningful visualizations and a dashboard.

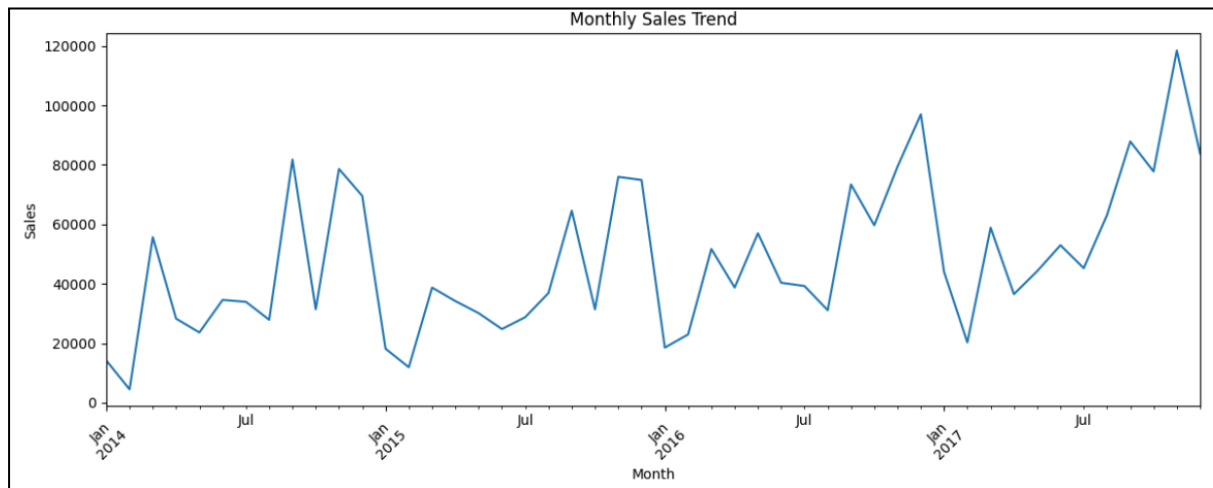
Project Overview:

This project focuses on analyzing the Superstore retail sales dataset to understand business and sales performance. The dataset contains information related to orders, customers, products, sales, profit, categories, sub-categories, and geographical regions. Python-based data analysis and visualization techniques are used to clean, explore, and analyze the dataset. Various charts are created to identify sales trends, category-wise performance, regional performance, and other important patterns. The final insights are presented through a dashboard that provides a summarized view of the business performance and helps in making data-driven decisions.

2. Data Analysis and Visualization

2.1 MONTHLY SALES TREND

Figure 1: Monthly Sales Trend



Interpretation:

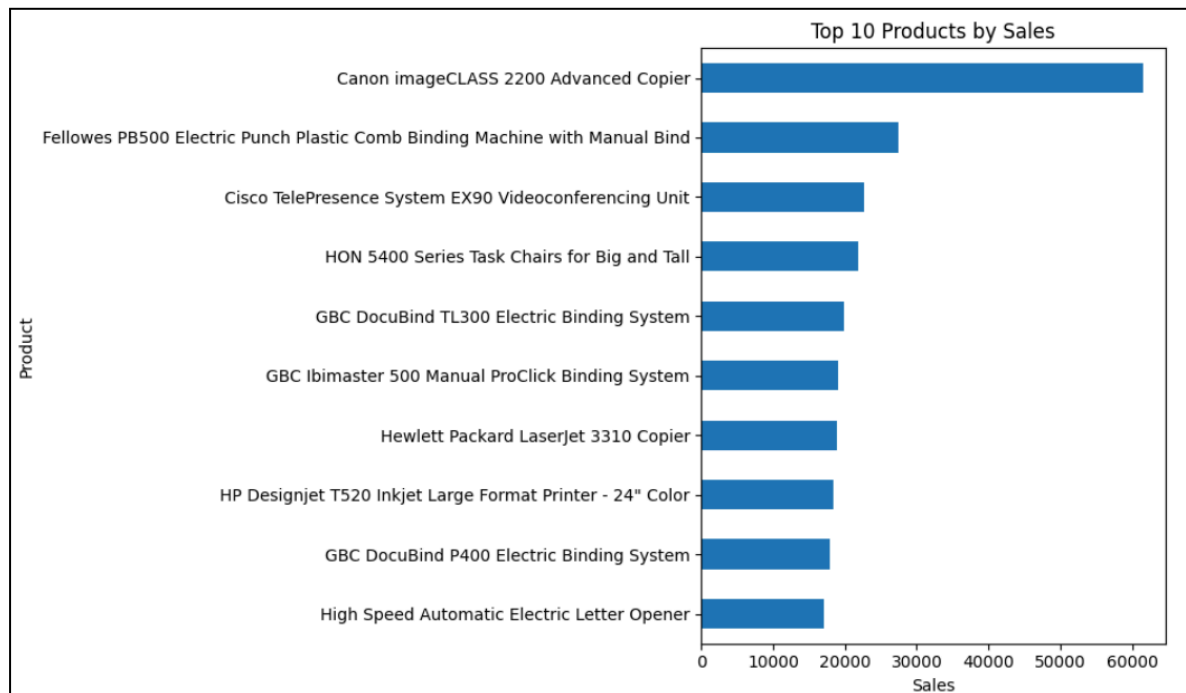
The monthly sales analysis shows considerable fluctuations in sales over the observed period. Several months show strong sales peaks, while other months record significantly lower sales. The highest visible sales peak occurs toward the end of the observed period, reaching approximately 110,000.

Business Insight:

The variation in monthly sales indicates that business performance is not consistent throughout the year. Identifying the factors associated with high- and low-performing periods can help the business improve sales planning and manage inventory more effectively.

2.2 Top 10 Products by Sales

Figure 2: Top 10 Products by Sales



Interpretation:

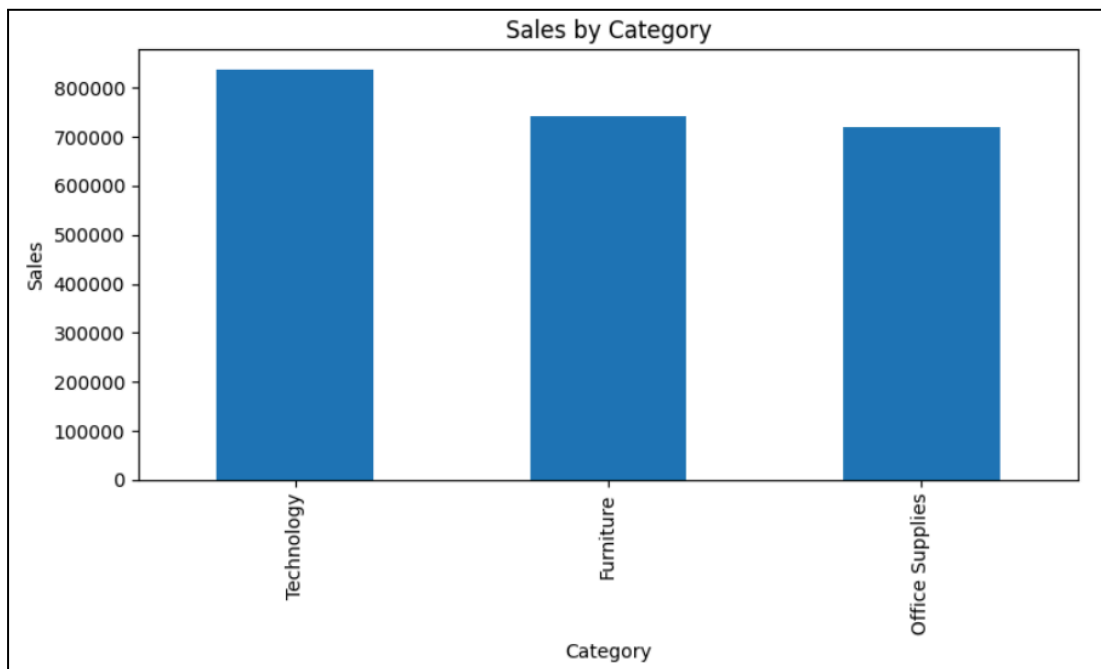
The analysis of the top-selling products shows that the Canon imageCLASS 2200 Advanced Copier is the leading product by sales among the top 10 products. Other high-performing products include printing, copying, binding, office equipment, and communication products.

Business Insight:

The strong performance of the leading products indicates that certain office technology and equipment products contribute significantly to sales. These products should receive appropriate attention in terms of inventory availability and sales planning.

2.3 Sales by Category

Figure 3: Sales by Category



Interpretation:

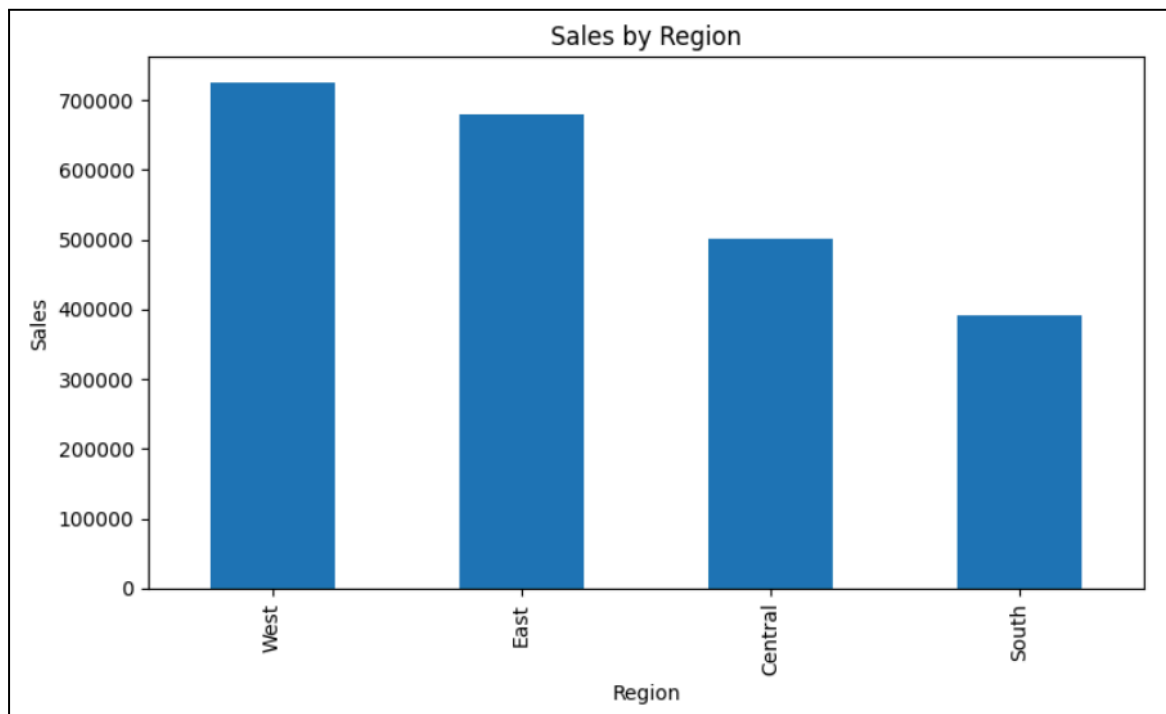
The category analysis shows that Technology is the highest-performing category, with sales of approximately 820,000. Furniture follows with approximately 670,000, while Office Supplies contributes approximately 590,000 in sales.

Business Insight:

Technology is the strongest revenue-generating category in the dataset. This suggests that the business should continue monitoring the performance of technology products and identify opportunities to expand or strengthen this category.

2.4 Sales by Region

Figure 4: Sales by Region



Interpretation:

The regional analysis shows that the West region has the highest sales, at approximately 720,000, followed by the East region at approximately 650,000. The Central region records approximately 450,000, while the South region has the lowest sales at approximately 310,000.

Business Insight:

The results show noticeable differences in sales performance across regions. The West and East regions are the strongest-performing markets, while the South and Central regions provide potential opportunities for improvement.

3. Key Business Insights

Based on the overall analysis, the following key business insights were identified:

1) Revenue Trend

Sales fluctuate significantly over time, with several high-performing and low-performing periods. This indicates the importance of monitoring sales patterns and planning according to demand.

2) Top Selling Products

The Canon imageCLASS 2200 Advanced Copier is the highest-selling product among the top 10 products analyzed.

3) High Value Categories

Technology is the highest-performing category, generating approximately 820,000 in sales, followed by Furniture and Office Supplies.

4) Regional Performance

The West region is the highest-performing region, followed by the East, Central and South. The South region has the lowest sales and therefore represents an area where additional analysis may be useful.

4. Business Recommendations

Based on the findings, the following recommendations can be considered:

4.1 Focus on High-Performing Products

The business should maintain adequate availability of top-selling products, particularly the Canon imageCLASS 2200 Advanced Copier and other high-performing office equipment.

4.2 Strengthen the Technology Category

Since Technology generates the highest category-level sales, the business should continue monitoring this category and identify opportunities for further growth.

4.3 Improve Lower-Performing Regions

The business should investigate the reasons for lower sales in the South and Central regions and identify opportunities for improving sales.

4.4 Monitor Monthly Sales Patterns

The business should analyze high- and low-sales periods to identify recurring patterns. This can help improve inventory and sales planning.

4.5 Use Regional and Product-Level Analysis Together

Combining regional and product analysis can help identify which products perform best in each market. This can support more targeted sales and marketing strategies.

5. Power BI Dashboard (Tool Used : Microsoft Power BI Desktop)

Dashboard Overview: The dashboard provides an interactive overview of sales performance using the Superstore dataset. It presents key business insights through visualizations such as total sales, total profit, sales trends over time, sales by category, and regional performance. The dashboard helps in understanding business performance and identifying important sales patterns.

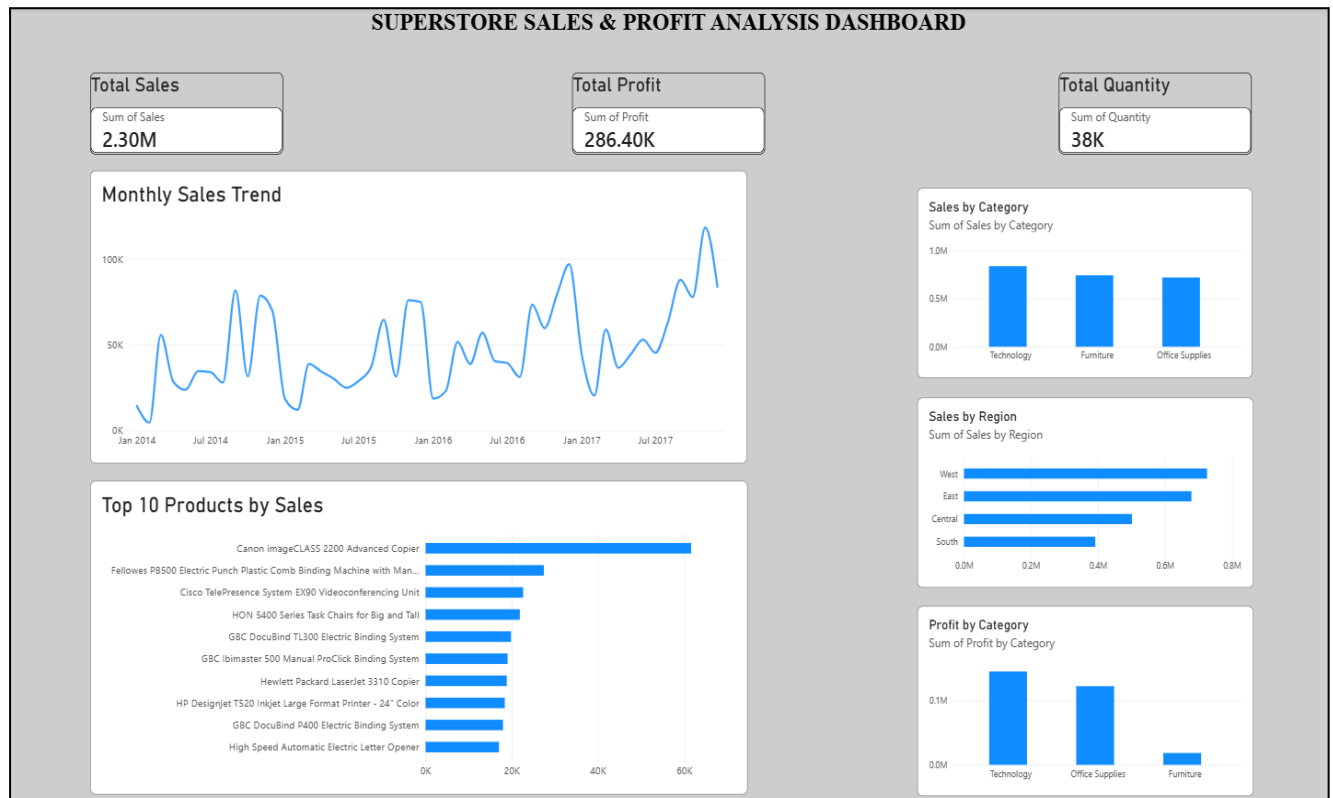


Figure: Superstore Sales & Analytics Dashboard

Key Insights:

- The dashboard provides a quick summary of overall sales and profit performance.
- The monthly sales trend helps identify changes in sales over time.
- Sales by category shows which product categories contribute the most to total sales.
- Regional analysis helps compare business performance across different regions.
- The dashboard makes the analysis easier to understand and supports data-driven business decisions.

6. Conclusion

This project successfully analyzed the Superstore retail sales dataset using data analysis and visualization techniques. The analysis provided a clear understanding of sales performance, profitability, product categories, customer segments, regional performance, and sales trends over time.

Data preprocessing and exploratory data analysis helped identify important patterns and relationships within the dataset. Various visualizations were used to present the findings in a simple and understandable manner. The final dashboard combines the major insights into a single view, making it easier to monitor business performance and interpret the results.

Overall, the project demonstrates how data analytics and visualization can be used to transform raw sales data into meaningful business insights. These insights can help businesses understand their performance, identify areas for improvement, and support better data-driven decision-making.